

Agile Project Management

CHAPTER 1: INTRODUCTION TO AGILE

Chapter Objectives

In this chapter, we will introduce:

- ◆ The Software Development Life Cycle (SDLC)
- ◆ The role of an architect and technical lead in software delivery

Chapter Concepts

Software Development Life Cycle

Agile Development

Chapter Summary

What Is SDLC?

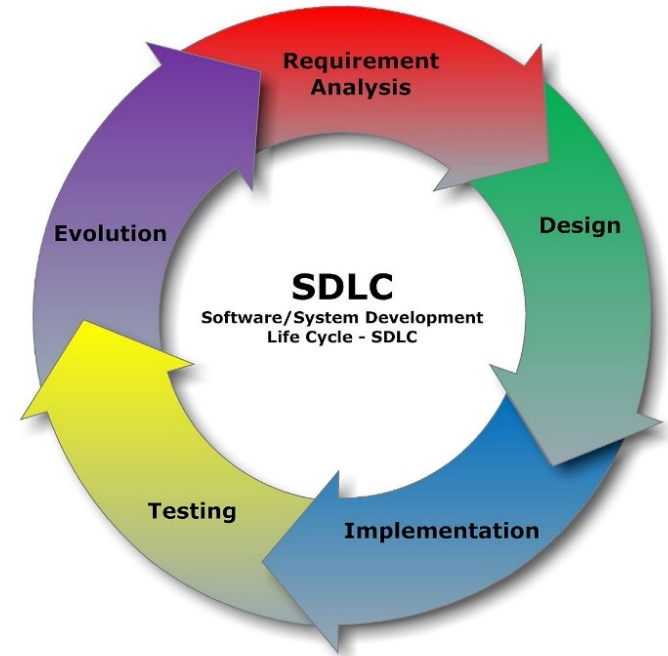
- ◆ Software Development Life Cycle (SDLC) is a process for developing, maintaining, replacing software
 - Aims to produce a high-quality software that:
 - ◆ Meets or exceeds customer expectations
 - ◆ Reaches completion within times and cost estimates

SDLC Models

- ◆ Following are the most important and popular SDLC models followed in the industry:
 - Waterfall Model
 - Iterative Model
 - Spiral Model
 - V-Model
 - Big Bang Model
- ◆ There is also a consideration of the role and makeup of teams
 - Distinct teams/departments
 - ◆ E.g., Development, test/QA, operations/support
 - DevOps
 - ◆ Those that build it, deploy, and support it

Stages of SDLC

- All models have a number of key stages shown on the diagram
- The main difference between models is the size of the work units performed on each cycle around the loop
- Focus and key tasks to be performed depend on project management approach as well as the type and size of the project



Agile Projects

- ◆ Using Scrum/Kanban, XP Agile principles and processes
- ◆ Focus around self-organizing teams with lightweight processes and frequent delivery
 - Align all architecture work with the culture of development teams
 - Deliver often, in small iterations
 - Give more power to the teams
 - Favor individuals and interactions over processes and tools, working software over comprehensive documentation
- ◆ Agile != Chaos
 - Architecture is still very important
 - ◆ Done in iterative fashion as well
 - Long term planning and roadmaps are useful
 - ◆ Changing direction by 180° is expensive

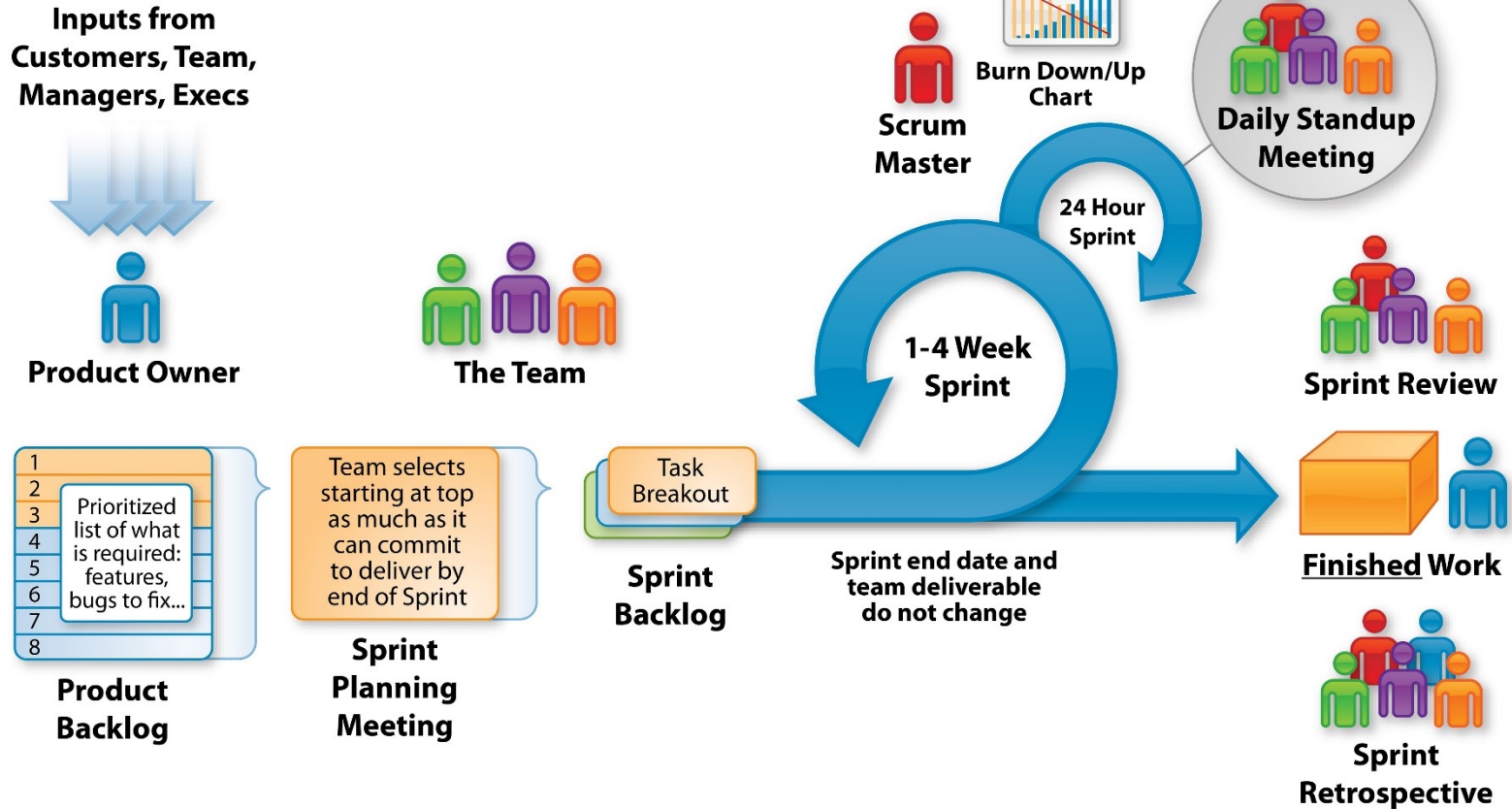
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The Agile Scrum Framework at a glance



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Overview of Scrum Process

- Starting point is list of all things system should include and address
 - Functionality
 - Features
 - Technology
- Work is undertaken by Scrum teams
 - Small, cross-functional teams
 - Take on part of product backlog to turn into increment in allocated days
 - Allocated day period is known as *Sprint*
- Multiple teams can develop increments in parallel
 - Scrum of Scrums

Agile Framework Collaborators

- **Product Owner** maintains all product requirements, features, etc. in **Product Backlog**
- **Scrum Team** selects a subset of items from Product Backlog (**Sprint Backlog**) based on priority and other factors and implements them in iterative fashion, through **Sprints**
 - A time box of one month or less during which a “Done”, useable, and potentially releasable **‘product increment’** is created
- During Sprint
 - Work effort is planned and coordinated through short, 15-minute **Daily Scrum (Standup)** meetings
 - **Scrum Master** facilitates activities, removes obstacles, and protects the team from outside interruptions and distractions
- At the end of the Sprint:
 - Scrum team demonstrates product to key stakeholders during **Sprint Review** meeting
 - Reflects on ways to improve during **Sprint Retrospective** meeting

Meet the Team

- ◆ Cross-functional team responsible for *all* aspects of development
 - Analysis
 - Design
 - Implementation
 - Testing
- ◆ Balance of team member skills is vital in achieving successful outcomes
 - Technical skills
 - Domain knowledge
 - Communication skills
- ◆ Commits to achieving Sprint goal
 - Has full authority to do whatever is necessary to achieve the goal
 - Only constraints are organizational standards and conventions



Product Backlog

Evolving, prioritized queue of business and technical functionality that needs to be developed into a system

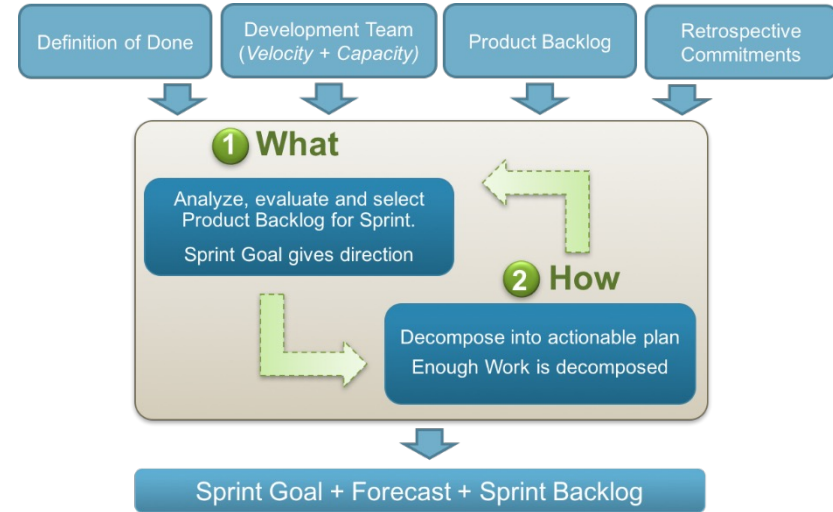
- Product Owner maintains product backlog based on requirements from various stakeholders
 - Dev Team and Scrum Master can contribute to the backlog
- List of:
 - Features
 - Functions
 - Technologies
 - Enhancements
 - Bug fixes
 - Anything that represents 'work to be done'
- Backlog is dynamic
 - Requirements emerge/evolve
 - Exists as long as product exists

Estimating Product Backlog Effort

- ◆ Scrum team estimates the effort of the items in the backlog
- ◆ Estimating is an iterative process
 - Estimates change as items become better understood
 - Initial estimate can be done as part of 'backlog grooming'
- ◆ Estimate is not binding
 - It is a starting point—best guess
 - Continually refined
- ◆ Product Owner is prioritizing backlog items
 - Ultimate decision maker in terms of what is important
 - ◆ Team can provide feedback and suggest
- ◆ Includes time to:
 - Define architecture, design, test, and build

Planning a Sprint

- ◆ The work to be performed in the Sprint is planned at the Sprint Planning
 - This plan is created by the collaborative work of the entire Scrum team
 - Meeting is normally 8 hours or less
- ◆ During Sprint Planning, Scrum team decides on:
 - **What** to deliver in the next Sprint
 - **How** will the work be achieved
- ◆ Two key outcomes of Sprint Planning:
 - Sprint Goal (the 'what')
 - ◆ A short statement of what the team plans to achieve during the sprint
 - Sprint Backlog (the 'how')
 - ◆ List of the product backlog items the team commits to delivering
 - ◆ Breakdown of the backlog items into tasks, with more detailed estimates



Sprint Constraints

- Every product development is constrained by four variables
 - Time
 - Cost (people and resources)
 - Delivered quality
 - Delivered functionality
- During Sprint, the first three are fixed
 - Fourth can be changed
 - ◆ As long as it still meets the Sprint goal

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